devices, which often lack access to good sources of entropy or even reliable clocks that could be used for deterministic nonces. Pressure to reduce the size of messages may also result in developers using nonces that are too small to be randomly generated safely. An attacker may also be able to influence conditions to make nonce reuse more likely, such as by tampering with the clock, or exploiting weaknesses in network protocols, as occurred in the KRACK attacks against WPA2 (https://www.krackattacks.com). In the worst case, where a nonce is reused for many messages, the algorithms in NaCl and COSE both fail catastrophically, enabling an attacker to recover a lot of information about the encrypted data and in some cases to tamper with that data or construct forgeries.

To avoid this problem cryptographers have developed new modes of operation for ciphers that are much more resistant to accidental or malicious nonce reuse. These modes of operation achieve a security goal called *misuse-resistant authenticated encryption* (MRAE). The most well-known such algorithm is AES-SIV, based on a mode of operation known as *Synthetic Initialization Vector* (SIV, https://tools.ietf.org/html/rfc5297). In normal use with unique nonces, SIV mode provides the same guarantees as any other authenticated encryption cipher. But if a nonce is reused, a MRAE mode doesn't fail as catastrophically: an attacker would only be able to tell if the exact same message had been encrypted with the same key and nonce. No loss of authenticity or integrity occurs at all. This makes AES-SIV and other MRAE modes much safer to use in environments where it might be hard to guarantee unique nonces, such as IoT devices.